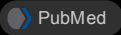
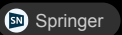
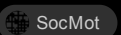
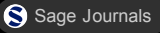


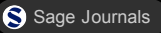
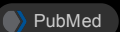
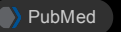
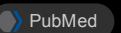
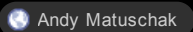
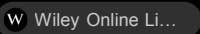
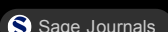
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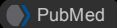
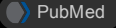
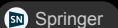
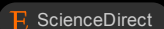
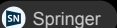
Yes. If you want to understand the machinery behind **TikTok-level engagement but redirect it toward learning and skill development**, these are the 25 papers I would put in the WayPoint curriculum.

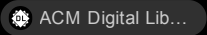
I'm ranking them by usefulness to what we're building, not simply citation count. Some are foundational psychology; others are learning science, game design, recommender systems, or warnings about persuasive technology.

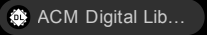
1. **Ryan & Deci (2000) — “Self-Determination Theory and the Facilitation of Intrinsic Motivation, Social Development, and Well-Being.”** *American Psychologist*. This is foundational. Learn autonomy, competence, and relatedness. For WayPoint, the big question is how progression can make users *want to become competent* rather than merely chase points. 
2. **Schultz, Dayan & Montague (1997) — “A Neural Substrate of Prediction and Reward.”** *Science*. The classic reward-prediction-error paper. Essential for understanding anticipation, unexpected rewards, learning from outcomes, and why predictable rewards behave differently from surprising ones. Do **not** reduce this to the internet cliché “dopamine = addiction.” The actual mechanism is much more useful. 
3. **Wood & Neal (2007) — “A New Look at Habits and the Habit–Goal Interface.”** *Psychological Review*. Explains how behaviors move from deliberate actions toward cue-triggered habits. Directly relevant to turning “I should practice” into “this is just what I do after lunch/on the train/before bed.” 
4. **Oulasvirta et al. (2012) — “Habits Make Smartphone Use More Pervasive.”** *Personal and Ubiquitous Computing*. Extremely relevant to TikTok-style products. The researchers identify short, repetitive **checking habits** around dynamic smartphone content. Study this alongside Wood & Neal. 
5. **Loewenstein (1994) — “The Psychology of Curiosity: A Review and Reinterpretation.”** *Psychological Bulletin*. The famous **information-gap** account of curiosity: awareness of a gap between what you know and what you want to know creates motivation to close it. This is one of the scientific foundations for good hooks. 
6. **Gollwitzer (1999) — “Implementation Intentions: Strong Effects of Simple Plans.”** *American Psychologist*. The science behind “When X happens, I will do Y.” Useful for getting learning out of the app and into someone's real life: “After morning formation, spend five minutes reviewing X.” 
7. **Kivetz, Urminsky & Zheng (2006) — “The Goal-Gradient Hypothesis Resurrected.”** *Journal of Marketing Research*. People increase effort as they perceive themselves getting closer to a goal. This is the science you should

understand before designing XP bars, skill levels, quests, course completion indicators, and near-mastery states. 

8. **Berger & Milkman (2012)** — “**What Makes Online Content Viral?**” *Journal of Marketing Research*. Examines which content characteristics are associated with sharing. Particularly useful for understanding emotional arousal, usefulness, interestingness, and why some ideas propagate while others die. Apply it to educational content hooks, not clickbait. 
9. **Roediger & Karpicke (2006)** — “**Test-Enhanced Learning: Taking Memory Tests Improves Long-Term Retention.**” *Psychological Science*. Probably one of the most important papers for WayPoint. Retrieving information improves later retention substantially compared with simply rereading it. Your feed therefore shouldn't just show information; it should constantly make people **recall** information. 
10. **Cepeda et al. (2006)** — “**Distributed Practice in Verbal Recall Tasks: A Review and Quantitative Synthesis.**” *Psychological Bulletin*. Major quantitative review of the **spacing effect**. It provides the scientific foundation for resurfacing something tomorrow, next week, and next month instead of treating learning as a one-and-done feed item. 
11. **Karpicke & Blunt (2011)** — “**Retrieval Practice Produces More Learning than Elaborative Studying with Concept Mapping.**” *Science*. Reinforces just how powerful retrieval can be, including for meaningful learning rather than simple memorization. WayPoint should optimize for “produce an answer” rather than “I recognize that answer.” 
12. **Sweller (1988)** — “**Cognitive Load During Problem Solving: Effects on Learning.**” *Cognitive Science*. Foundational cognitive-load theory. Critical for microlearning because reducing content to 30 seconds doesn't automatically make it cognitively efficient. You need to understand working-memory limitations and schema acquisition. 
13. **Chi et al. (1994)** — “**Eliciting Self-Explanations Improves Understanding.**” *Cognitive Science*. Learners improve understanding by explaining material to themselves rather than merely receiving explanations. That suggests a powerful WayPoint interaction: “**Explain why you chose that answer.**” 
14. **Taylor & Rohrer (2010)** — “**The Effects of Interleaved Practice.**” *Applied Cognitive Psychology*. Instead of completing 20 nearly identical questions in succession, mix related problem types so learners must determine *which method applies*. Particularly important for practical troubleshooting and technical training. 
15. **Hattie & Timperley (2007)** — “**The Power of Feedback.**” *Review of Educational Research*. Essential for understanding when feedback helps and what useful feedback should actually communicate. This matters enormously if WayPoint evaluates answers, simulations, scenarios, or physical skills. 

16. **Dunlosky et al. (2013)** — “Improving Students’ Learning With Effective Learning Techniques.” *Psychological Science in the Public Interest*. A major review comparing common learning techniques. Read this early because it will stop us from wasting time gamifying weak learning methods. Practice testing and distributed practice come out particularly well. 
17. **Moneta & Csikszentmihalyi (1996)** — “The Effect of Perceived Challenges and Skills on the Quality of Subjective Experience.” *Journal of Personality*. Important empirical work around **flow**, challenge, and skill. This points toward dynamically adjusting question/task difficulty so users stay near the boundary of their ability. 
18. **Ryan, Rigby & Przybylski (2006)** — “The Motivational Pull of Video Games: A Self-Determination Theory Approach.” *Motivation and Emotion*. Very relevant. It connects game enjoyment and persistence with psychological need satisfaction. This is a better foundation for game mechanics than simply copying badges, leaderboards, and streaks. 
19. **Sailer et al. (2017)** — “How Gamification Motivates: An Experimental Study of the Effects of Specific Game Design Elements on Psychological Need Satisfaction.” *Computers in Human Behavior*. Instead of asking whether “gamification works,” it asks what specific game elements do psychologically. That’s exactly how we should evaluate XP, avatars, badges, progress indicators, competition, teams, etc. 
20. **Corbett & Anderson (1994)** — “Knowledge Tracing: Modeling the Acquisition of Procedural Knowledge.” *User Modeling and User-Adapted Interaction*. This is where WayPoint starts becoming technically interesting. **Knowledge tracing** estimates what a learner knows from their interaction history. It is basically the foundation for maintaining a continuously changing internal skill model. 
21. **Piech et al. (2015)** — “Deep Knowledge Tracing.” *NeurIPS*. Applies recurrent neural networks to sequences of student interactions to estimate evolving knowledge. You don’t need this architecture in V1, but you absolutely need to understand the problem it is solving. 
22. **Covington, Adams & Sargin (2016)** — “Deep Neural Networks for YouTube Recommendations.” *ACM RecSys*. One of the papers I specifically want you to read for the **TikTok/YouTube side** of this project. It explains a production-scale recommender architecture and separates candidate generation from ranking. Think about replacing “what will they watch?” with “what learning activity should they receive next?” 
23. **Li, Chu, Langford & Schapire (2010)** — “A Contextual-Bandit Approach to Personalized News Article Recommendation.” *WWW* . This introduces a very useful concept for WayPoint: **exploration vs. exploitation**. Sometimes recommend

what you're confident the learner needs; sometimes deliberately explore something new to learn more about their interests and ability. 

24. Fogg (2009) — “A Behavior Model for Persuasive Design.” *Persuasive*. Fogg's model proposes that behavior occurs when motivation, ability, and a prompt converge. It's deliberately simple, but very useful when diagnosing product friction: “They want to learn but starting requires too much effort,” for example. 

25. Mathur et al. (2019) — “Dark Patterns at Scale: Findings from a Crawl of 11K Shopping Websites.” *CSCW*. Non-negotiable because we're deliberately studying persuasive design. The paper analyzes thousands of real-world interfaces and develops a taxonomy of manipulative design patterns. If WayPoint learns how to influence behavior, the team also needs to recognize when influence becomes coercion, deception, or exploitation. 

The first papers I would actually read are **#1, #2, #5, #9, #10, #12, #18, #20, #22, and #25**. That combination gives you motivation, reward, curiosity, memory, cognitive load, games, learner modeling, recommendation systems, and the ethical boundary.

There is also a useful distinction I want us to keep throughout this research:

TikTok's optimization problem

Choose next item $\rightarrow \max P(\text{continued engagement})$

WayPoint's optimization problem

Choose next activity $\rightarrow \max (\text{learning gain} + \text{retention} + \text{motivation} + \text{skill pr}$
subject to something like:

$\text{cognitive load} \leq \text{manageable}$

$\text{difficulty} \approx \text{current skill} + \epsilon$

$\text{user agency and wellbeing} > \text{minimum acceptable threshold}$

That difference should eventually show up **inside the recommendation objective itself**, not just in our marketing language.

And I would go one step further. As we read these papers, we should create a **WayPoint Behavioral Design Bible**. For every paper, extract: psychological mechanism $\rightarrow$ evidence $\rightarrow$ product mechanic $\rightarrow$ metric $\rightarrow$ potential abuse $\rightarrow$ WayPoint implementation. By paper 25 we'd have the beginnings of the actual behavioral architecture for the game/app, rather than 25 sets of notes sitting in a folder.

